

Cryptography & Cyber Law

Course code: ICT 4115

1. Abstract Algebra & Number Theory fundamental:

(i) Fermat's Little Theorem (FLT):

If p is a prime number and $p \nmid a$, then

$$a^{p-1} \equiv 1 \pmod{p}$$

Proof Sketch: The non-zero elements of $\mathbb{Z}_p$ form a multiplicative group of order $p-1$. By Lagrange's Theorem, the order of any element divides the order of the group.

$$\text{hence, } a^{p-1} \equiv 1 \pmod{p}$$

Miller-Rabin Primality Test:

When $n-1 = 2^s \cdot d$, where d is odd, for
a base $a \in [2, n-2]$:

(i) compute $x = a^d \pmod{n}$

(ii) If $x \equiv 1 \pmod{n}$ or $x \equiv -1 \pmod{n}$
then n passes.

(iii) Otherwise, square x up to $s-1$

times: $x \leftarrow x^2 \pmod{n}$. If $x \equiv -1 \pmod{n}$ at any steps n passes.

(iv) If $-1 \pmod{n}$ is never encountered
 n is definitely composite.

Evaluating:

$$n = 561 \quad (561 = 3 * 11 * 17):$$

$$21 \text{ } n=16=560=2^4 \cdot 35 \Rightarrow s=4, d=35$$

Testing with base $a=2$:

$$\bullet 2^{35} \equiv 263 \pmod{561} \text{ (neither 1 nor } -1)$$

$$\bullet 263^2 \equiv 166 \pmod{561}$$

$$\bullet 166^2 \equiv 67 \pmod{561}$$

$$\bullet 67^2 \equiv 1 \pmod{561} \neq -1$$

Since 1 was reached without immediately being preceded by -1 , 561 is composite by Miller-Rabin.

However, evaluating simple FLT gives

$$2^{560} \equiv 1 \pmod{561} \text{, because}$$

$$a^{n-1} \equiv 1 \pmod{n} \text{ holds for every } a \text{ coprime to } n$$

$$a^{560} \equiv 1 \pmod{561}$$

$$(a, 561) = 1 \text{ for every } a$$

So, 561 is a Carmichael number.

2. $\mathbb{Z}_n$ is a field if and only if n is prime

If n is prime

for any non-zero element $a \in \mathbb{Z}_p$, $\gcd(a, p) = 1$. By Bezout's identity, there exist integers x, y such that $ax + py = 1$ (mod p).

Thus, every non-zero element has a multiplicative inverse, making $(\mathbb{Z}_p, +, \cdot)$ a field.

If n is composite:

n can be factored as $n = a \cdot b$ where $1 < a, b < n$. In $\mathbb{Z}_n$,

$$a \cdot b \equiv 0 \pmod{n} \text{ with } a \not\equiv 0 \pmod{n} \text{ and } b \not\equiv 0 \pmod{n}$$

Examples:

- $\mathbb{Z}_{11}$ is field, because 11 is prime
- $\mathbb{Z}_{12}$ is not a field because
 $3 \times 4 \equiv 0 \pmod{12}$ meaning
neither 3 nor 4 possesses an
inverse.

3. Construct of $\text{GF}(\mathbb{Z}^3)$

- Irreducible polynomial: $f(x)$
 $= x^3 + x + 1$ over $\text{GF}(2)$.
- Let α be a root of $f(x)$, so
 $\alpha^3 + \alpha + 1 = 0 \Rightarrow \alpha^3 = \alpha + 1$

Field Elements:

The 8 elements are polynomials
of degree < 3

$$\mathbb{GF}(2^3) = \{0, 1, \alpha, \alpha^2, \alpha+1, \alpha^2+\alpha, \alpha^2+1, \alpha^2+\alpha+1\}$$

Cyclic Multiplicative Group:

The multiplicative group $\mathbb{GF}(2^3) \setminus \{0\}$ has order $2^3 - 1 = 7$

- $\alpha^1 = \alpha$
- $\alpha^2 = \alpha^2$
- $\alpha^3 = \alpha^2 + 1$
- $\alpha^4 = \alpha^2 + \alpha$
- $\alpha^5 = \alpha^3 + \alpha^2 = \alpha^2 + \alpha + 1$
- $\alpha^6 = \alpha^3 + \alpha^2 + \alpha = \alpha^2 + 1$
- $\alpha^7 = \alpha^3 + \alpha = 1$

Because $\alpha^k \neq 1$ for $1 \leq k < 7$, α has order 7. Thus, $\mathbb{GF}(2^3)^*$ is cyclic & generated by α .
 $\alpha^i \cdot \alpha^j = \alpha^{(i+j) \bmod 7}$

4. Square and multiply complexity

For an exponent e , its binary representation contains $k = \lceil \log_2 e \rceil + 1$ bits.

(i) Number of squaring : $\lceil \log_2 e \rceil$

(ii) " " multiplication : $\text{weight of } e$

(iii) Total time complexity : $O(\log e)$

Evaluating $7^{560} \pmod{561}$:

Since 561 is a Carmichael number and $\gcd(7, 561) = 1$:

$$7^{560} \equiv 1 \pmod{561}$$

∴ Result = 1